
DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Experiment No.: 1

Interfacing LPC2148 to LCD and display message on LCD

Name: _____

Roll No.: _____ *Batch:* _____

Date of Performance: _____

Date of Assessment: _____

Particulars	Marks
Marks for Regularity (05)	
Marks for Performance (15)	
Understanding (Viva) (05)	
Total (25)	
Signature of Course Teacher	

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Experiment: 1

AIM: Interfacing LPC2148 to LCD and display message on LCD.

OBJECTIVE:

1. To understand the architecture and pin configuration of a 16*2 LCD Module.
2. To configure LPC2148 GPIO pins as outputs for LCD control and data lines.
3. To write, compile and flash an Embedded C program for LCD display.

APPARATUS:

1. LPC2148 module development board.
2. Connecting wire.
3. Adapter.

THEORY:

LCD stands for Liquid Crystal Display. It is a flat panel display technology, mainly used in TVs and computer monitors, nowadays it is used for mobile phones also. In LCD, each pixel consists of a layer of molecules aligned between two transparent electrodes and two polarizing filters, the axes of transmission perpendicular to each other. The LCD is finding widespread use by replacing the LEDs.

LCD Pin Description:

The LCD requires 3 control lines (RS, R/W, and EN) and 8(or 4) data lines. The number of data lines depends on the mode of operation. If operated in 8-bit mode then 8-bit data lines are required. And if the operation is in 4-bit mode then 4-bit data lines are required. The 8-bit mode is faster than the 4-bit mode. In 8-bit mode, LCD uses a total of 14 pins including 8 data lines, 3 control lines, and 3 power supply lines (V_{cc}, V_{ss} and V_{ee})

1. Power Supply: The LCD discussed here uses three power supply pins (V_{cc}, V_{ss}, and V_{ee}) V_{cc} and V_{ss} pins are used to provide +5V and ground respectively. The pin V_{ee} is used for controlling LCD contrast.

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2. Control Lines: There are three control lines in the LCD. These three are used to control the LCD operations. There are two very important registers inside the LCD: the command register and the data register. The RS (Register select) pin is used to select the register out of these two. If RS = 0 the command register is selected and the user is allowed to send the command to the LCD. If RS = 1, the data register is selected and the data sent by the user is displayed on the LCD.

R / W (Read/ Write) pin allows the user to read/write the information (data or code) to/ from the LCD.

R / W = 1 when reading and R/W=0 when the writing operation is performed.

Another control pin EN (Enable) is used to latch the data present on the data pins. A high-low signal is required to latch the data. The LCD interprets and executes the commands at the instant the EN line is brought low.

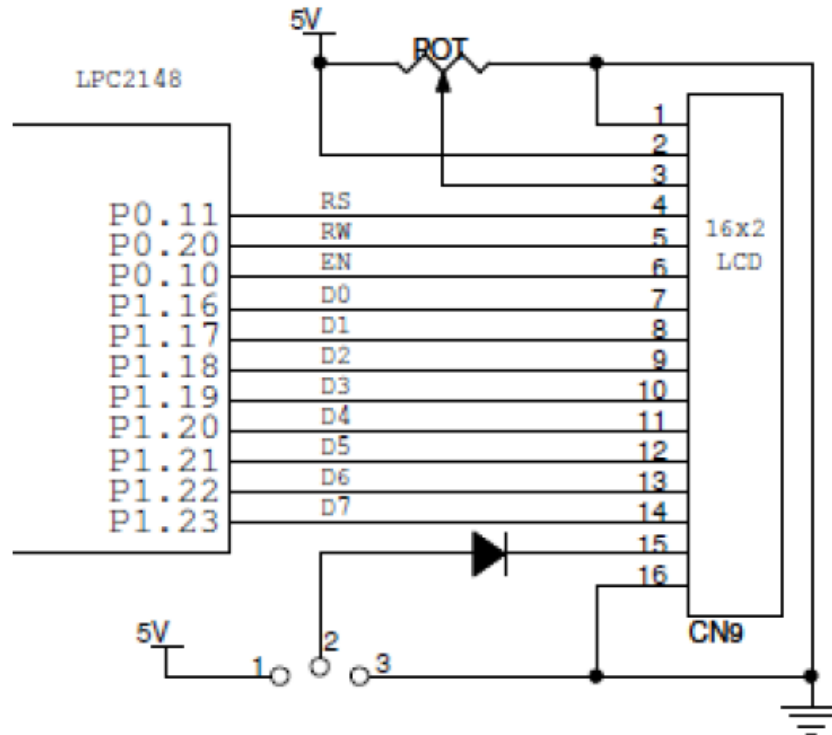
3. Data lines: The 8-bits data pins. D(0)-D(7) are used to send the information to the LCD or read the contents of the LCD's internal register.

PIN DESCRIPTION FOR LCD:

Pin No	Symbol	Description
1	Vss	Ground
2	Vcc	+5V power supply
3	Vee	power supply to control the contrast
4	RS	register select
5	R/W	Read/Write
6	EN	Enable
7-14	D0-D7	8- bit data lines

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INTERFACING DIAGRAM:



LCD INTERFACING DETAILS:

16x2 Character LCD	Pin Details	
	LPC2148	Description
D0	P1.16	LCD Data Bit 0
D1	P1.17	LCD Data Bit 1
D2	P1.18	LCD Data Bit 2
D3	P1.19	LCD Data Bit 3
D4	P1.20	LCD Data Bit 4
D5	P1.21	LCD Data Bit 5
D6	P1.22	LCD Data Bit 6
D7	P1.23	LCD Data Bit 7
EN	P0.10	LCD Enable Signal
RS	P0.11	LCD Register Select
RW	P0.20	LCD Read Write



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CONCLUSION:
